

# MoralBench + UniMoral

**No model has the highest saved score on all 8 tasks.**

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## MoralBench comparison

For every model pair, the two models' 95% score ranges overlap: 28 of 28 MFQ pairs and 45 of 45 vignette pairs.

## UniMoral model size and release date

Scores move up and down across tasks. Bigger or newer is not consistently higher here.

# MoralBench + UniMoral: no model has the top saved score on all 8 tasks

CEI local result. Same five models; each row keeps its own score type.

| Task                                                                     | Highest saved score | Score type       | Model                       |
|--------------------------------------------------------------------------|---------------------|------------------|-----------------------------|
| MFQ agreement<br>Track benchmark human ratings of value statements       | .884                | Agreement score  | <b>GPT-5.4 mini</b>         |
| Vignette agreement<br>Track benchmark human ratings of moral stories     | .920                | Agreement score  | <b>GPT-5.4</b>              |
| MFQ compare<br>Pick the higher-rated value statement                     | .550                | Accuracy         | <b>Haiku + Qwen tie†</b>    |
| Vignette compare<br>Pick the higher-rated moral story                    | .625                | Accuracy         | <b>Haiku + GPT-5.4 tie†</b> |
| UniMoral action<br>Match one recorded human choice                       | .668                | Accuracy         | <b>Claude Haiku 4.5</b>     |
| UniMoral typology<br>Classify the recorded choice into one of four types | .651                | Accuracy         | <b>Claude Opus 4.8</b>      |
| UniMoral factor<br>Pick a top-rated factor for the recorded choice       | .609                | Accuracy         | <b>Claude Haiku 4.5</b>     |
| UniMoral consequence<br>Write what could happen next                     | .152                | Text-match score | <b>Claude Opus 4.8</b>      |

† Every 95% range overlaps on both MoralBench comparison tasks. The apparent leaders are unresolved.

Each task uses its own score. Do not compare numbers from different rows.

# MoralBench: each pair of models has overlapping 95% score ranges

CEI local result. MFQ uses 8 models and 20 questions. Vignettes use 10 models and 24 questions.

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## MFQ value statements

**28 / 28**

model pairs have overlapping 95% score ranges

## Moral vignettes

**45 / 45**

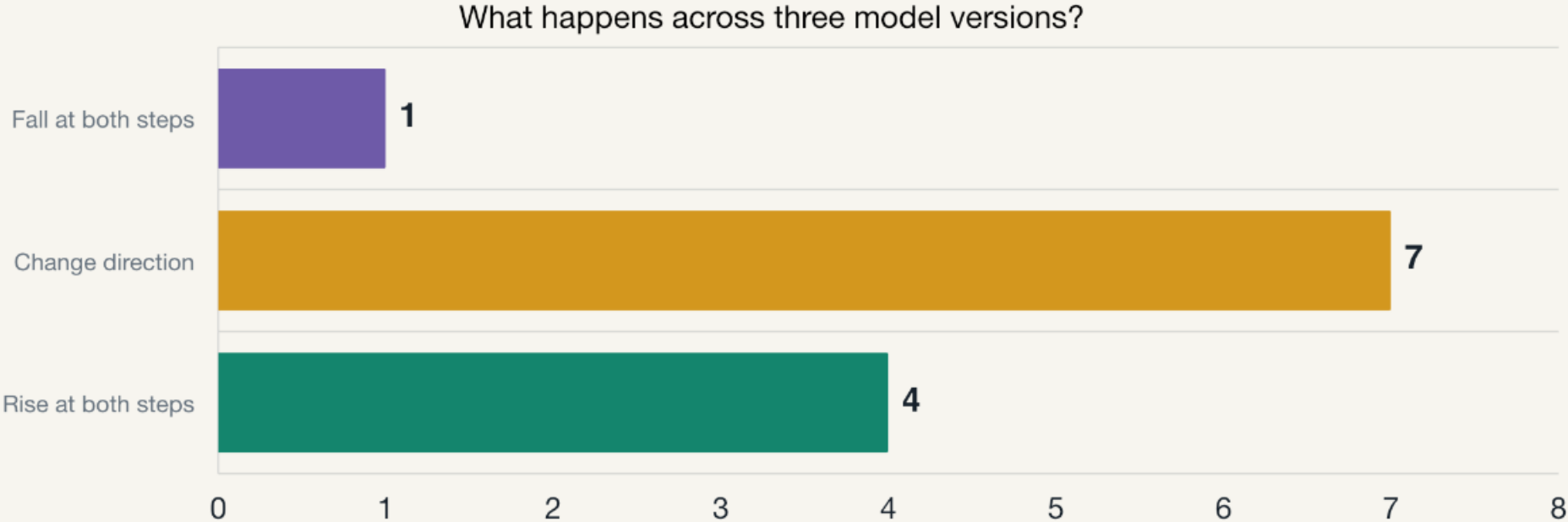
model pairs have overlapping 95% score ranges

**These saved results do not separate a leader.**

The ranges are not a direct model-pair test. Question-level scores and stronger uncertainty estimates are unavailable.

# UniMoral: only 4 of 12 family and task comparisons rise twice

Exploratory CEI result. Each comparison follows one model family on one task across three sizes.



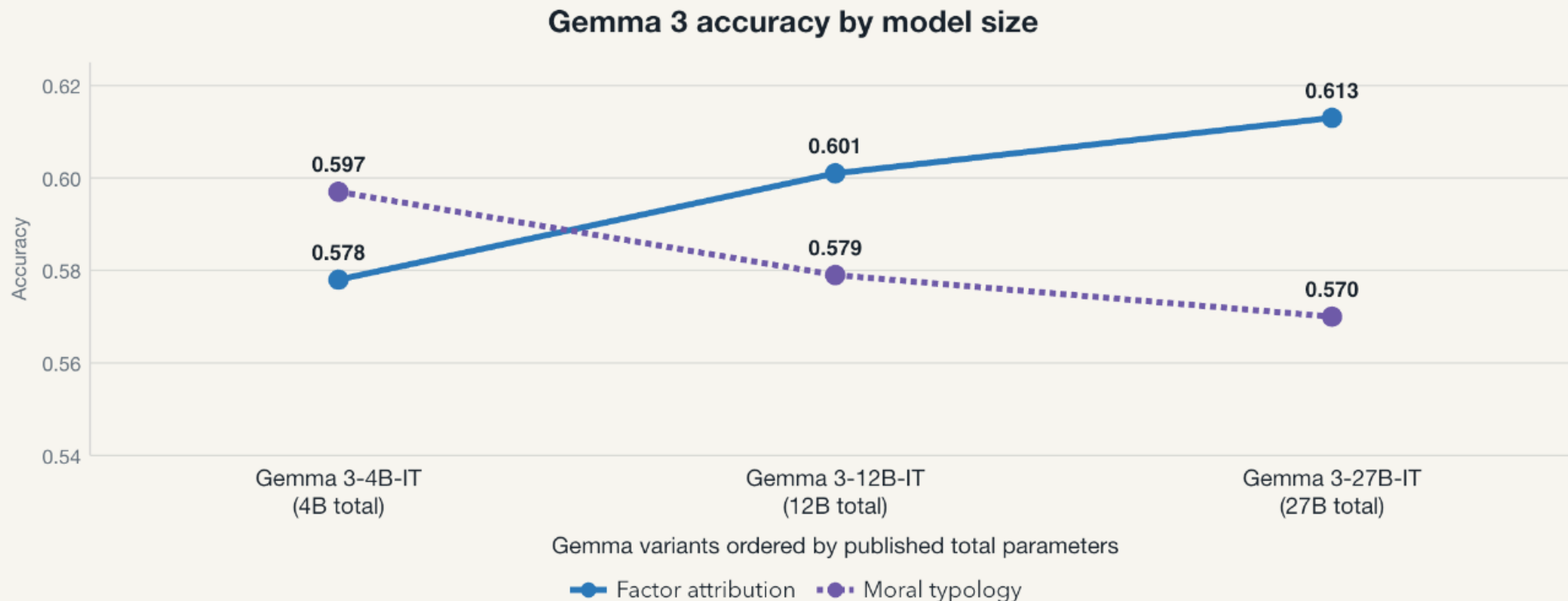
Each bar counts one model family on one task across three saved sizes.

**4 rise twice · 7 change direction · 1 falls twice**

This is exploratory. Model versions also differ in training and release, and no confidence ranges were saved.

# UniMoral Gemma: factor rises while typology falls

Exploratory CEI result. Gemma 3 at 4B, 12B, and 27B; the vertical axis is zoomed.



**4B to 27B: factor accuracy  $+0.034$  · moral typology accuracy  $-0.027$**

Each point is one saved task score. No uncertainty range or raw run archive is available.

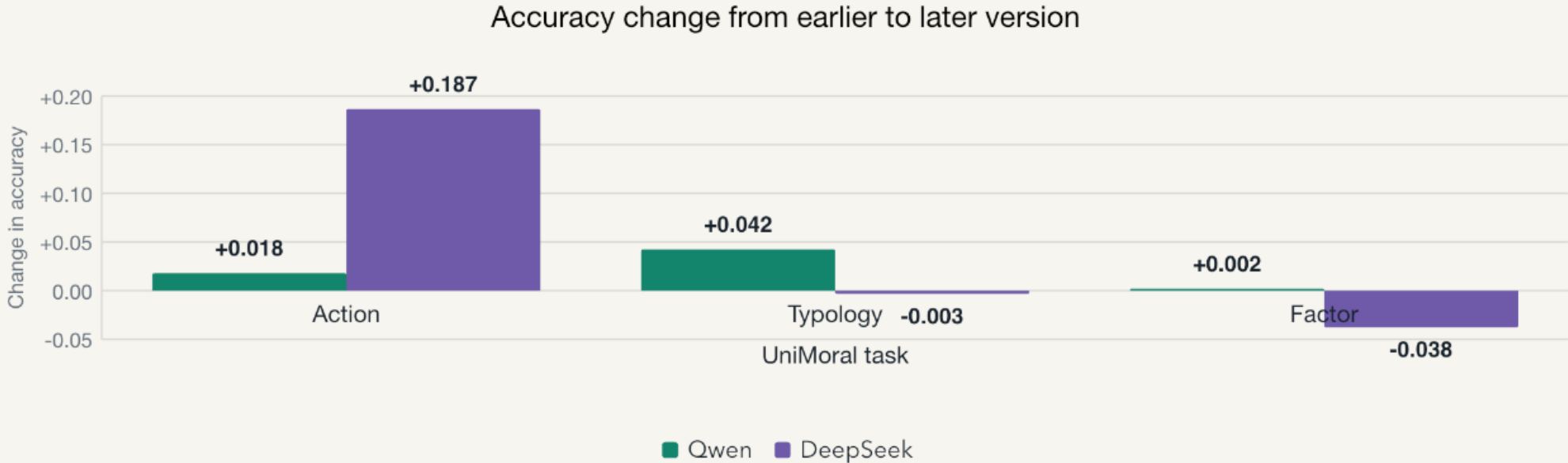
# UniMoral releases: Qwen rises on 3 tasks, DeepSeek on 2

Exploratory CEI result. Later score minus earlier score; consequence uses text match.

**Qwen:** Qwen2.5 7B Instruct · 2024-09-19 · 7.61B total to Qwen3.5 9B · 2026-02-27 · 9B total

**DeepSeek:** V3-0324 · 2025-03-24 · 671B main / 37B active to V4 Flash · 2026-04-22 · 284B main / 13B active

DeepSeek labels show published main-model B and active B used per token.



Across all 4 tasks: Qwen is higher on 3 and lower on 1. DeepSeek is higher on 2 and lower on 2.

Separate metric · Consequence text match (METEOR): Qwen  $-0.027$  · DeepSeek  $+0.019$

All scores were tested 28–29 May 2026. Release date describes the model, not the test date. No uncertainty ranges were saved.

# Five papers ask different questions. None is exactly repeated here.

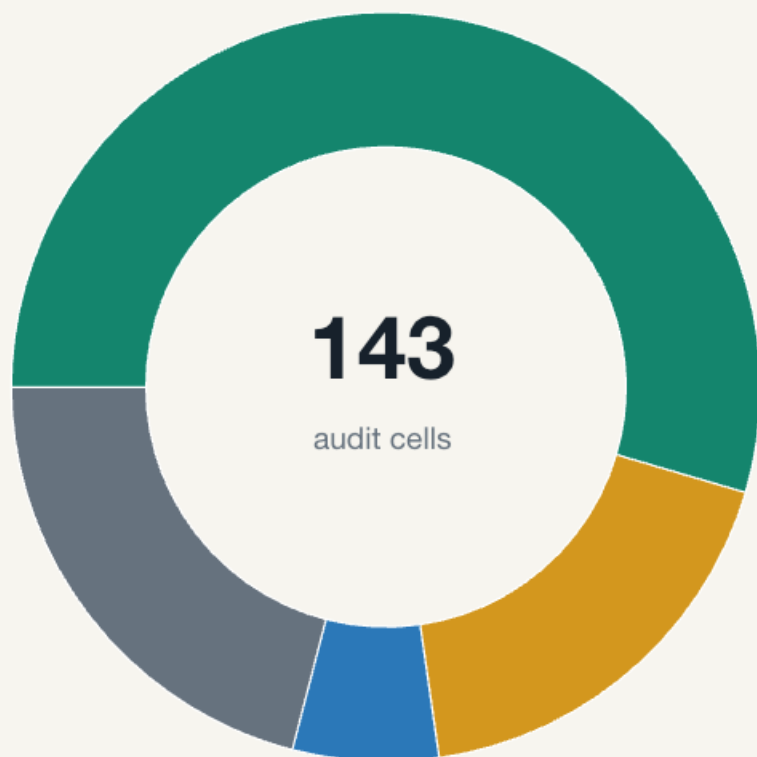
Paper results and CEI results use different models, data, prompts, or scores.

| Paper              | What it asks                                                  | Headline paper result                      | Closest CEI evidence            |
|--------------------|---------------------------------------------------------------|--------------------------------------------|---------------------------------|
| MoralBench         | Do model choices track human ratings?                         | Different models lead its four assessments | Related tasks, different setup  |
| UniMoral           | Can models predict actions, types, factors, and consequences? | Llama action F1 ranges from .506 to .662   | Narrower versions of four tasks |
| MoReBench          | Does reasoning meet expert criteria?                          | Harmless 81.1%; logical 47.9%              | Keyword check only              |
| MoralLens          | Does reasoning order change reasons and choices?              | CD GAP .230 to .052; utility .641 to .792  | Keyword check only              |
| Value Kaleidoscope | How do value tasks change with model size?                    | Most gains occur before 3B                 | Different models and scoring    |

Keep paper scores and CEI scores separate. They answer related but different questions.

# CEI evidence audit: 78 of 143 results support main findings

The other 65 results are checks, a separate multimodal extension, or exclusions.

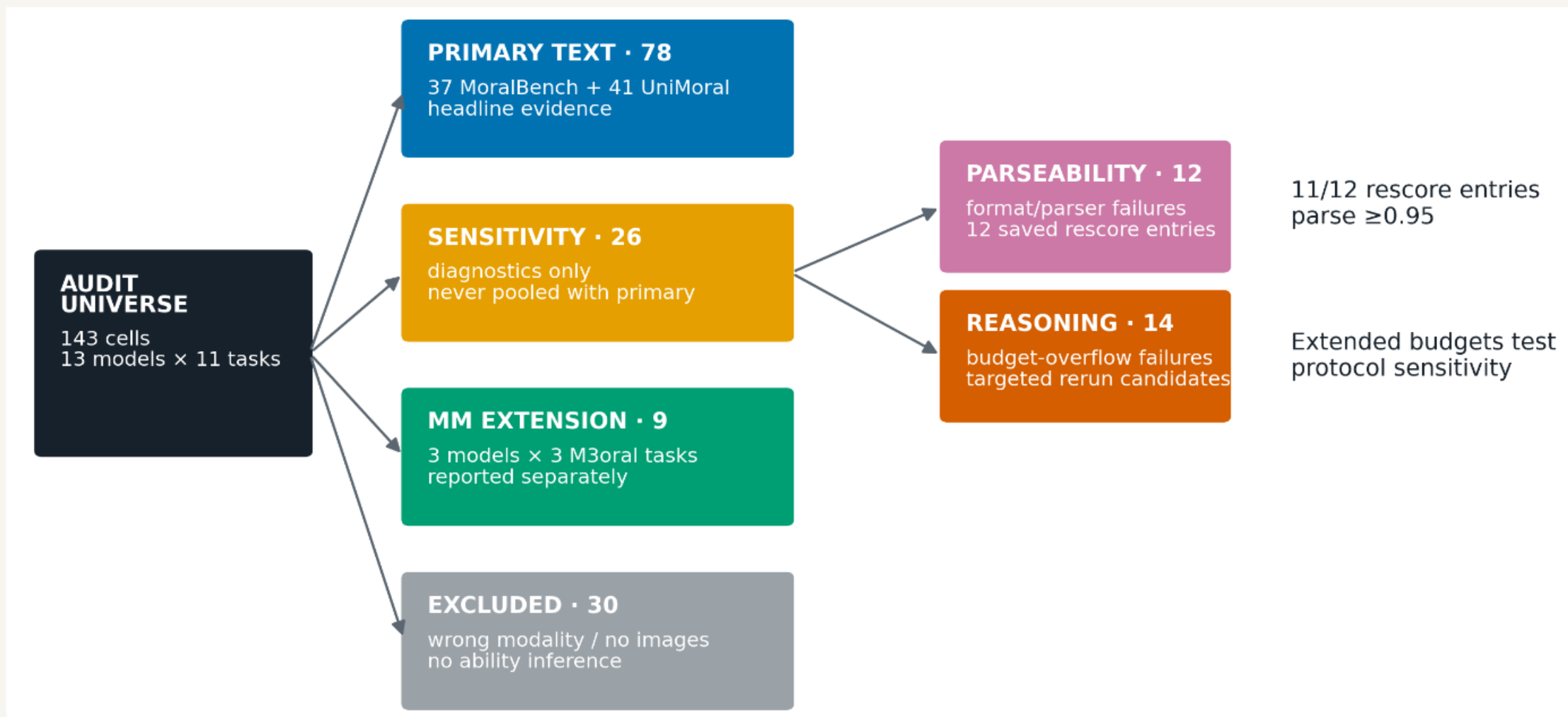


- **78 · Main text · 55%**  
Support the main task findings
- **26 · Checks only · 18%**  
Use to test robustness
- **9 · Multimodal · 6%**  
Report as a separate extension
- **30 · Excluded · 21%**  
Do not use for model claims

The other 65 results should not be used for headline model claims.

# CEI evidence audit: what each result group can support

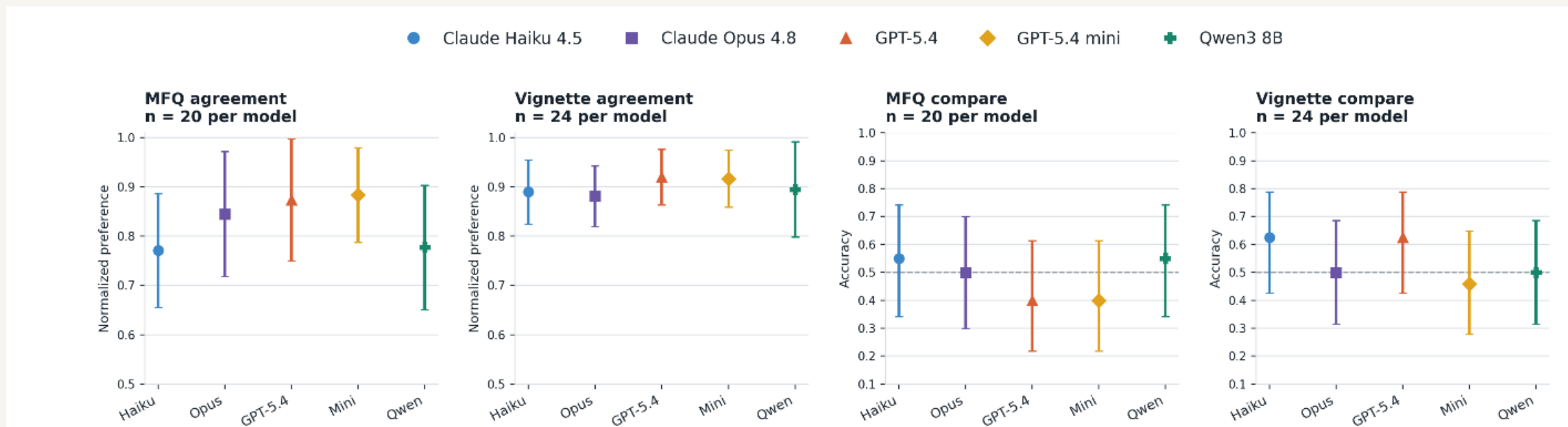
143 saved results are split into main findings, checks, multimodal results, and exclusions.



A file check confirms that the package is intact. It does not make weak or excluded evidence stronger.

# MoralBench: different models have the highest saved score

CEI local result. x = model; y = task score; dot = score; line = saved 95% range.



Agreement and accuracy stay separate. These ranges are not direct tests between two models.

# MoralBench: all model pairs have overlapping 95% ranges

CEI local result. 28 of 28 MFQ pairs and 45 of 45 vignette pairs overlap.

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## Moral Foundations (MFQ)

**100%**

**28 of 28 model pairs**

8 models · 20 questions

## Moral vignette

**100%**

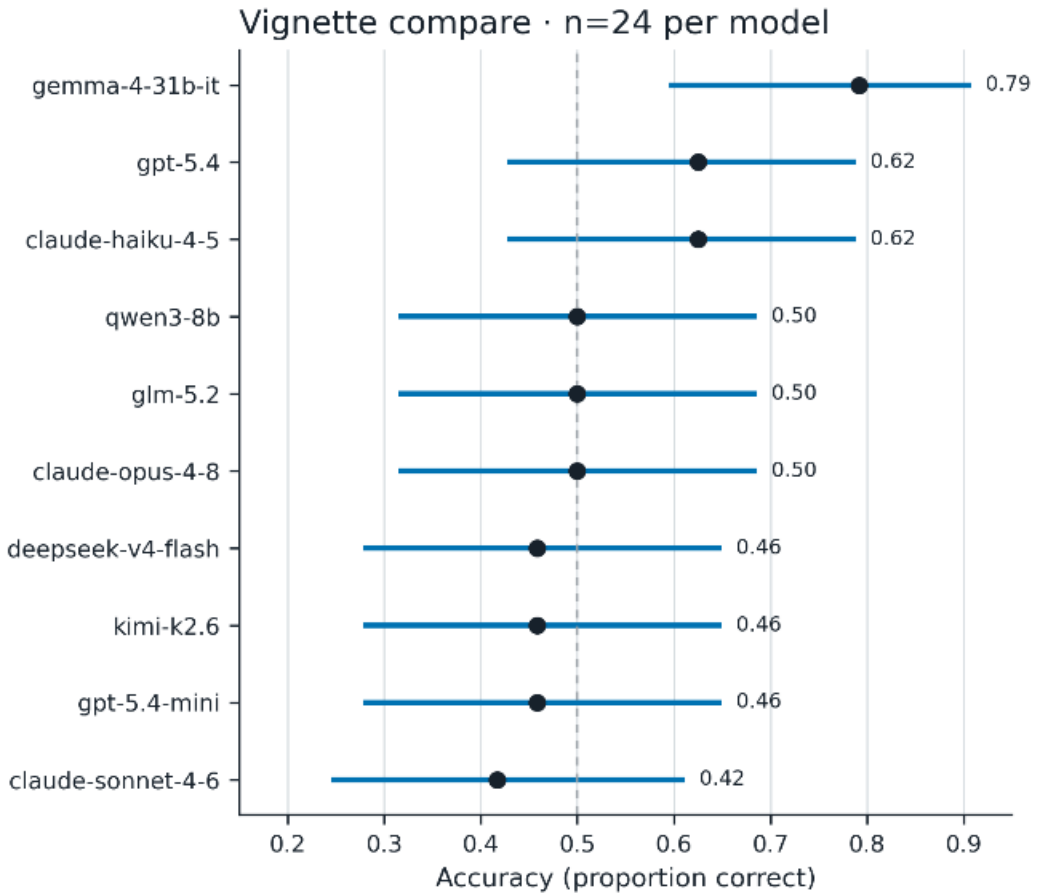
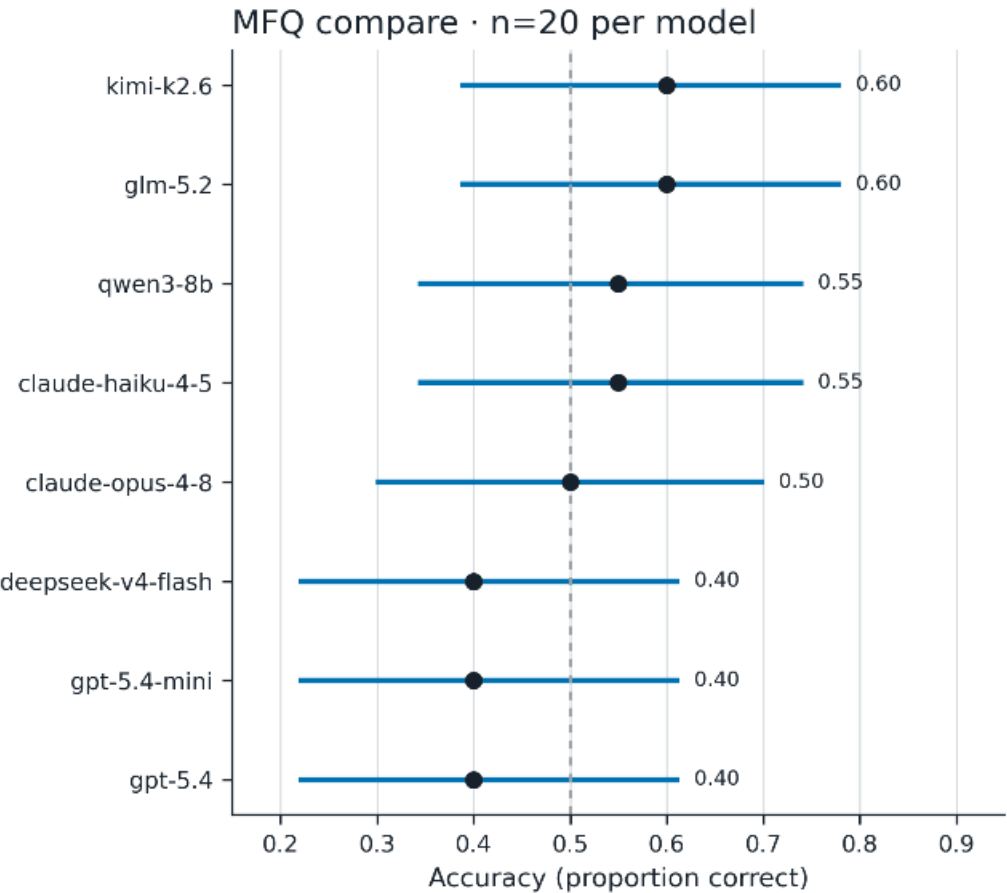
**45 of 45 model pairs**

10 models · 24 questions

This is not a direct model-pair test. Question-level scores and stronger uncertainty estimates are missing.

# MoralBench: all 18 saved intervals are wider than .30

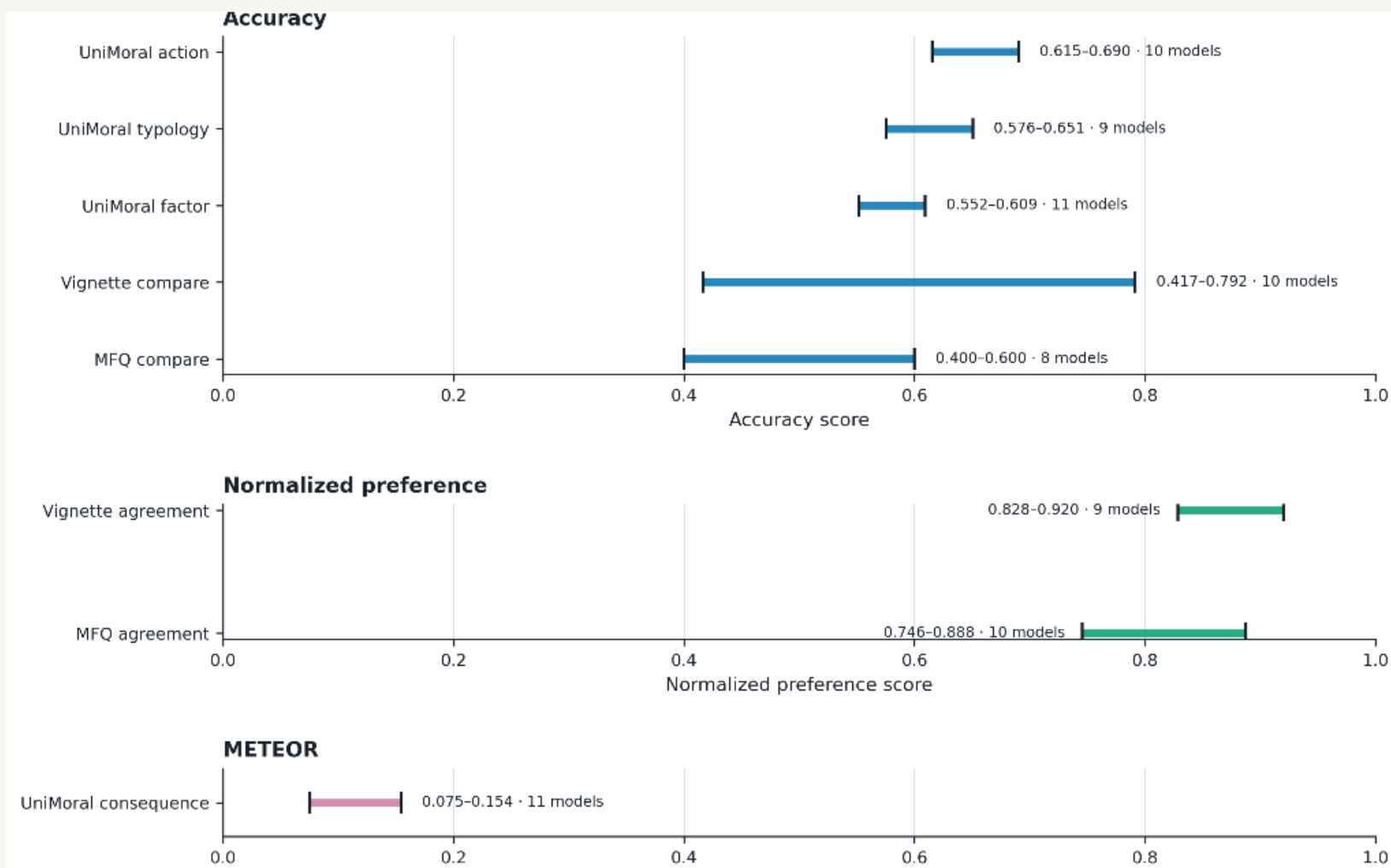
CEI local result. x = accuracy; y = model; dot = score; line = saved 95% range.



The calculation does not use paired model answers or related-question structure. It does not test human validity.

# MoralBench + UniMoral: keep the 3 score types separate

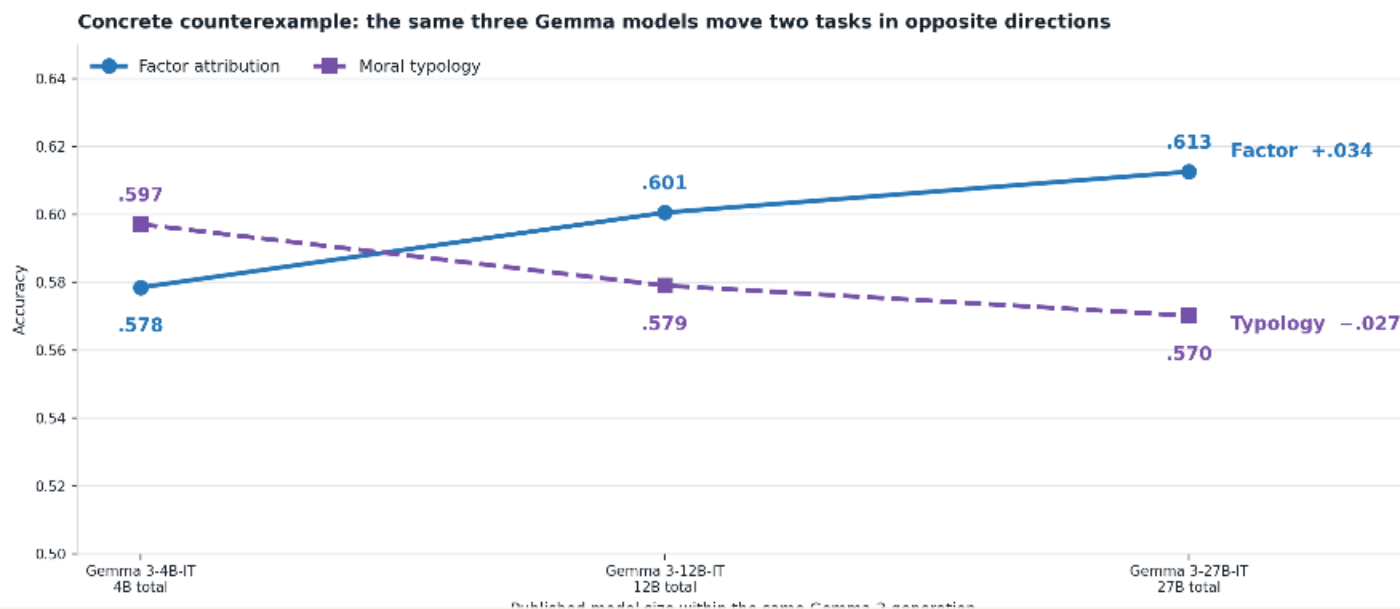
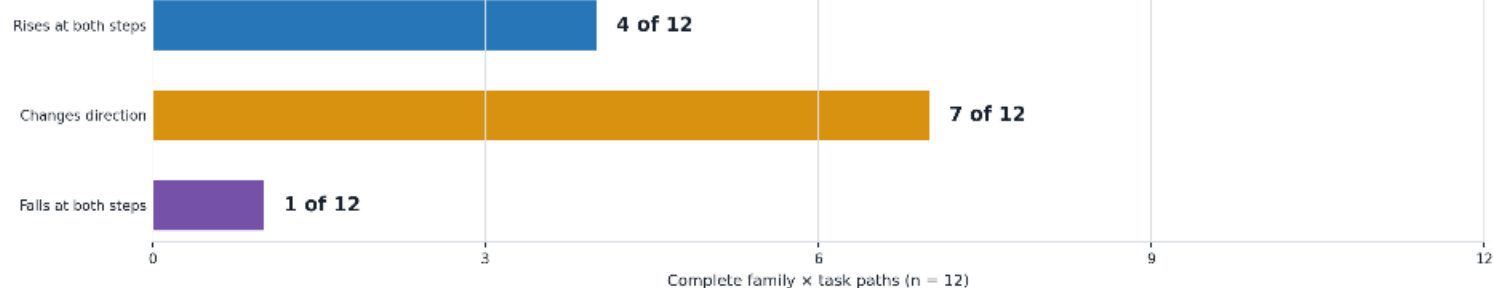
CEI local result. x = score within one metric; each line = one task's five-model range.



A high score in one panel cannot be compared with a high score in another. There is no overall moral score here.

# UniMoral: only 4 of 12 family and task comparisons rise twice

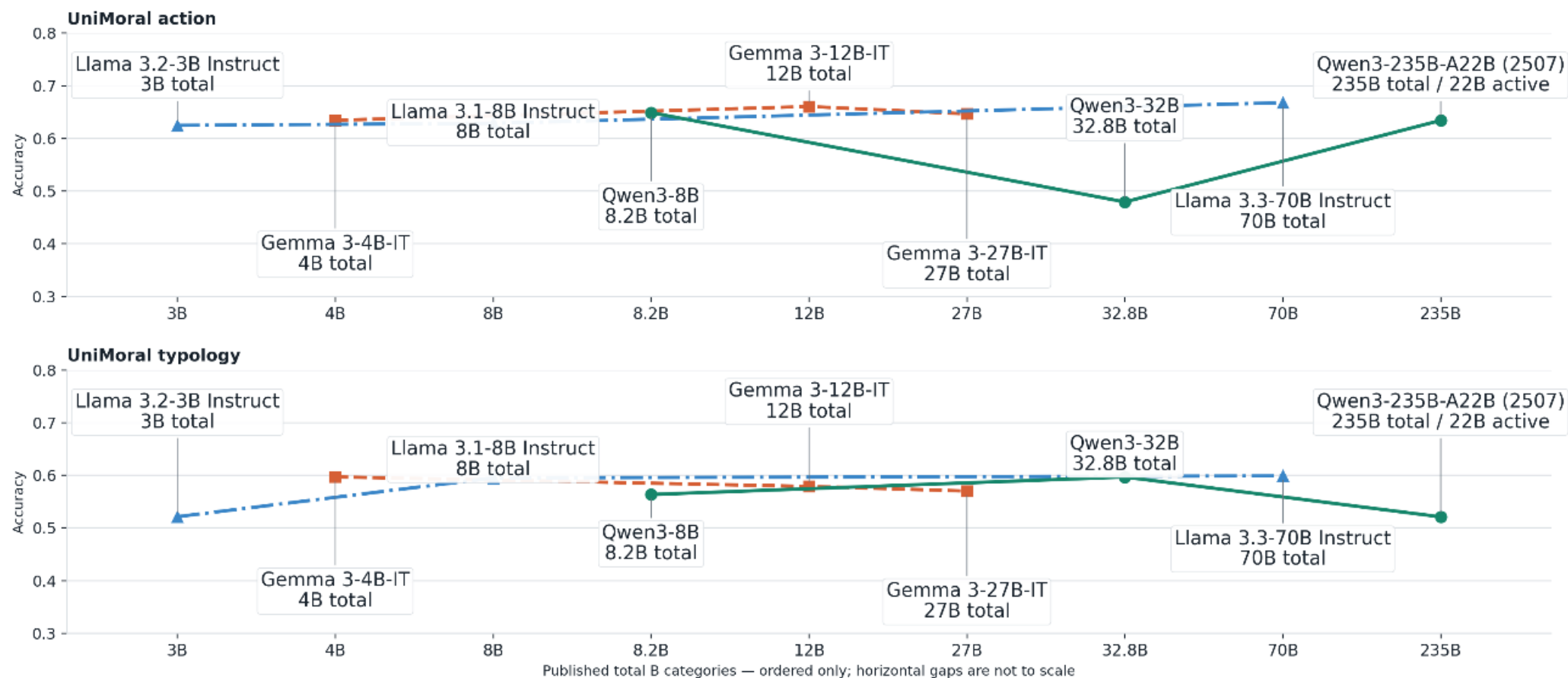
Exploratory CEI result. Bars = 12 comparisons; lines = Gemma at 4B, 12B, and 27B.



The Gemma vertical axis is zoomed. Size is not the only model difference. No uncertainty ranges were saved.

# UniMoral size: 3 of 6 family and task comparisons change direction

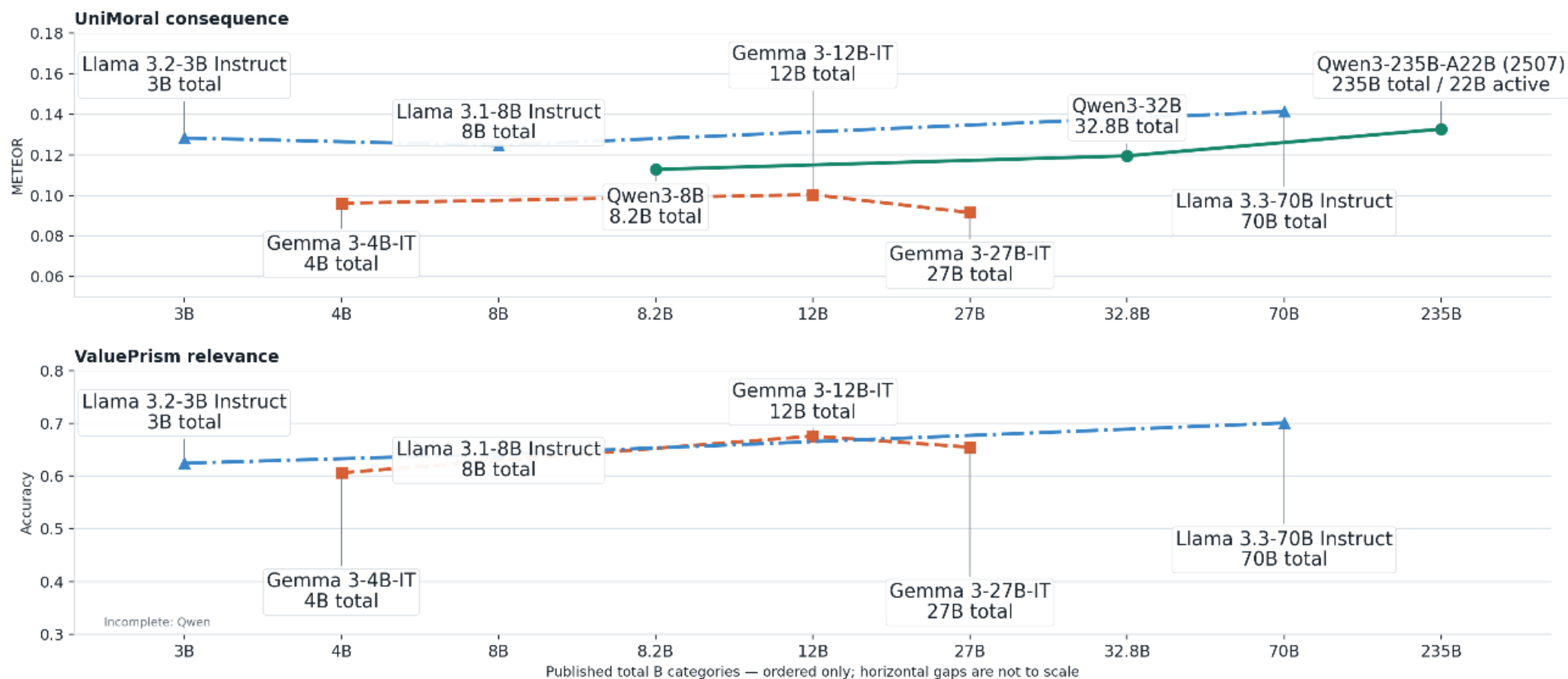
Exploratory CEI result. x = published size category, ordered only; y = accuracy; labels = model + B.



Each line stays within one family and task. No saved uncertainty ranges or raw run archive are available.

# UniMoral and ValuePrism size: 3 of 5 comparisons change direction

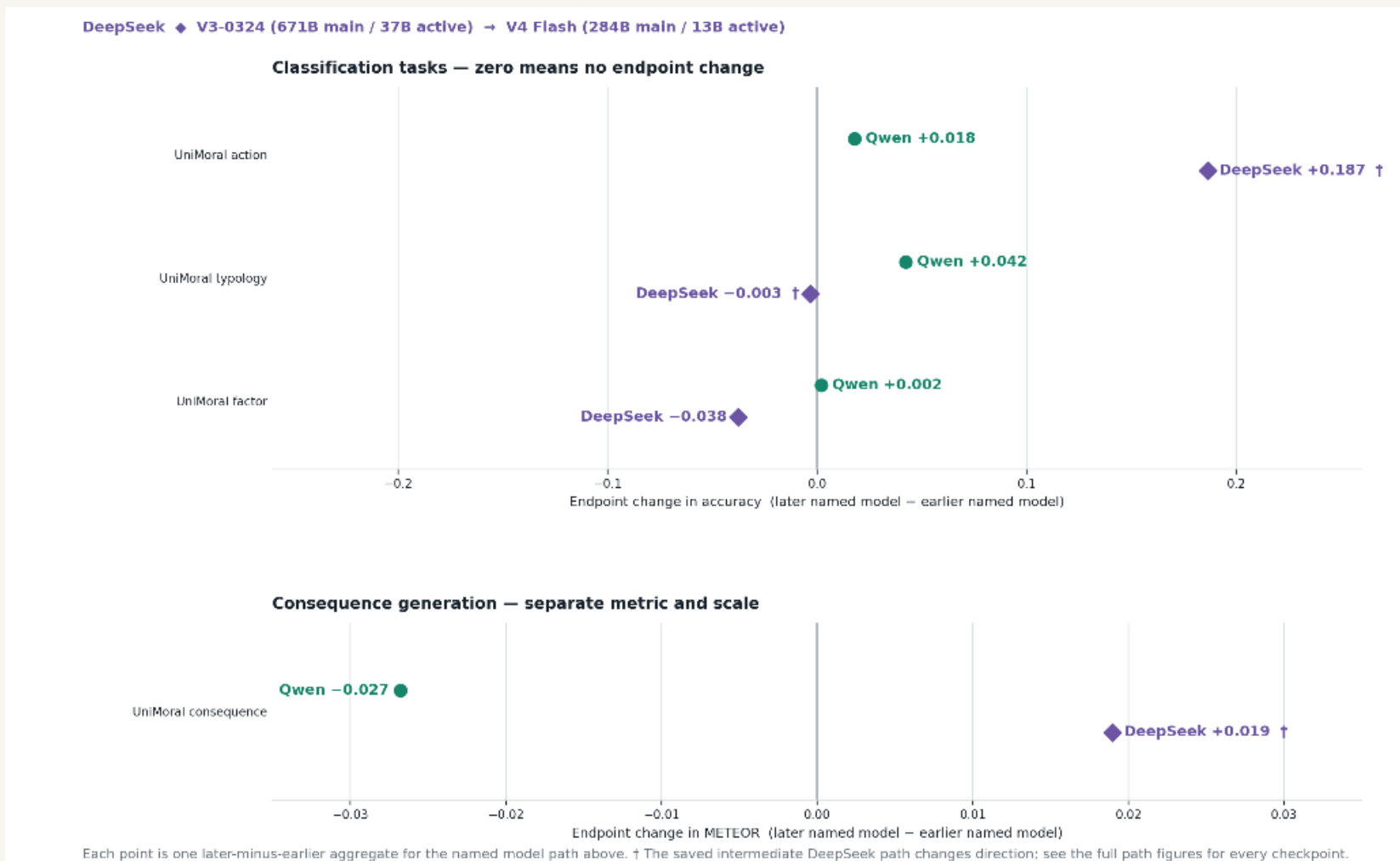
Exploratory CEI result. x = published size category, ordered only; y = named metric; labels = model + B.



The CEI ValuePrism test is not a repeat of the paper's Kaleido experiment. No uncertainty ranges were saved.

# UniMoral: Qwen is higher on 3 tasks; DeepSeek on 2

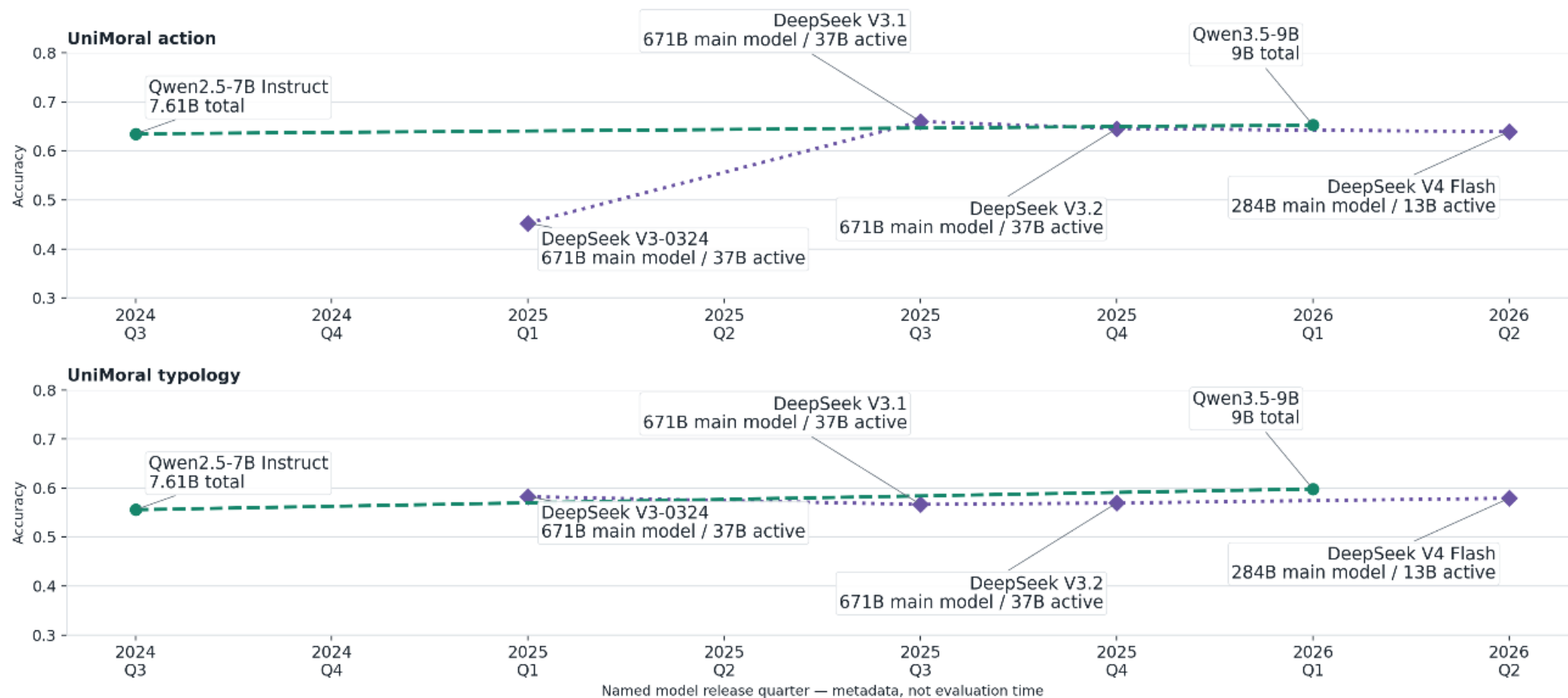
Exploratory CEI result.  $x$  = later minus earlier score;  $y$  = task; metrics stay separate.



All scores were tested on May 28 or 29, 2026. Release date describes the model, not the test date.

# DeepSeek reverses on UniMoral action and typology

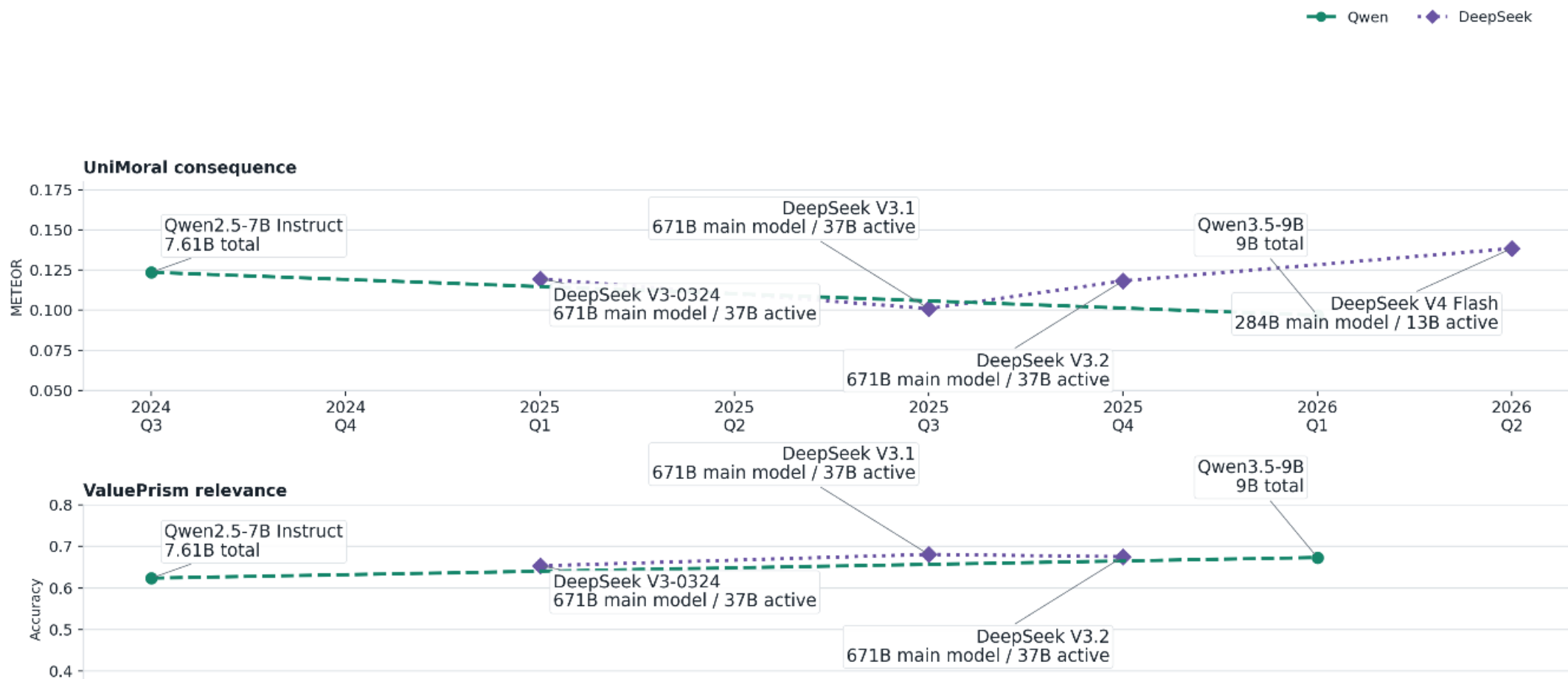
Exploratory CEI result. x = model release quarter; y = accuracy; labels = model + B.



Release quarter is model metadata, not test time. The lines do not show a cause. The available files have no saved uncertainty ranges or raw run archive.

# UniMoral and ValuePrism: later Qwen falls on consequence and rises on relevance

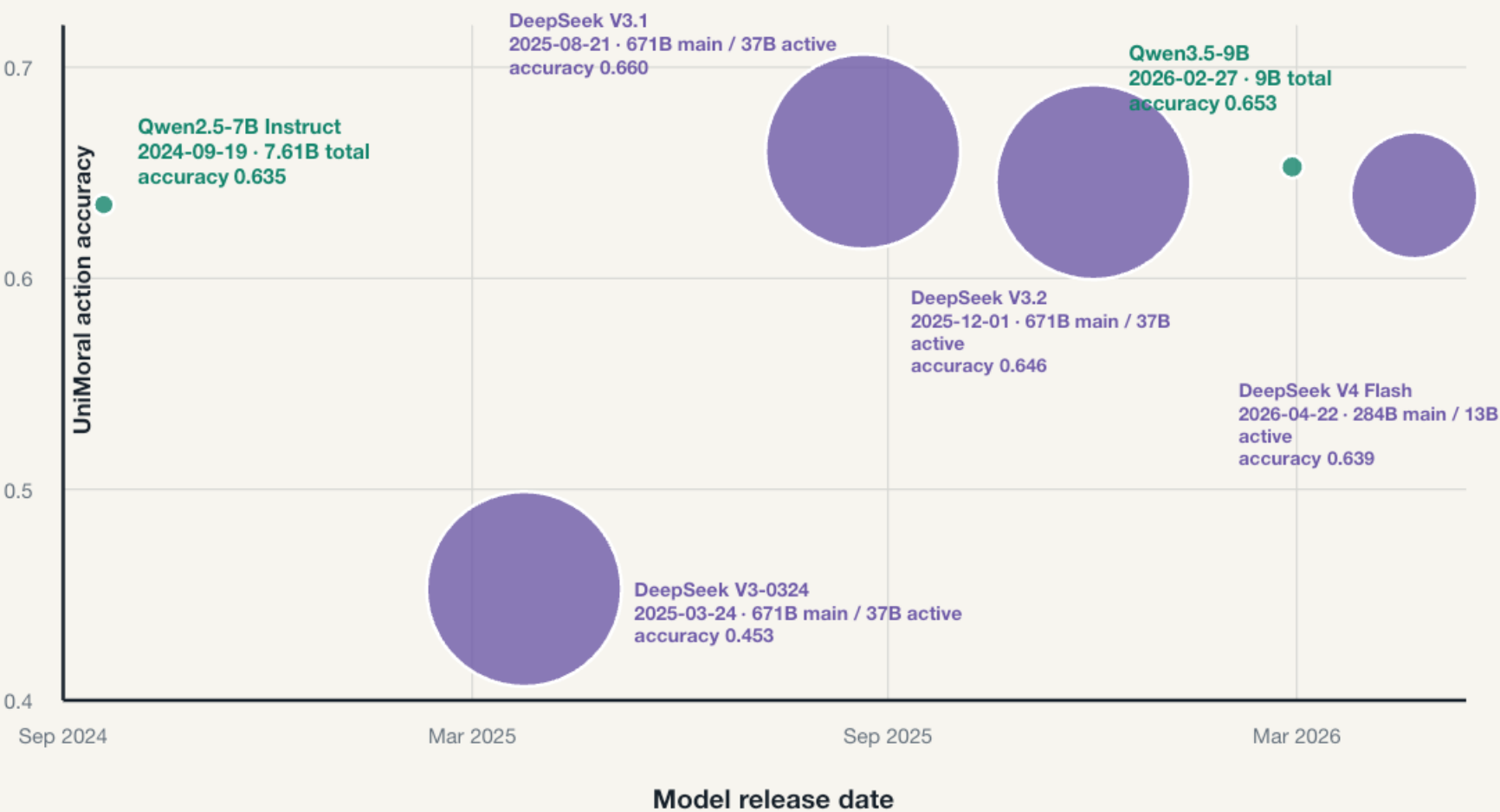
Exploratory CEI result. x = model release quarter; y = named metric; labels = model + B.



Consequence uses text match. Relevance uses accuracy. Release quarter is model metadata. The available files have no saved uncertainty ranges or raw run archive.

# UniMoral action: 671B DeepSeek releases range from .453 to .660

Every point names the model, release date, published size, and CEI accuracy.



Same published size, different scores

DeepSeek 671B

.453 .660 .646

These 6 points do not show a steady size or date pattern.

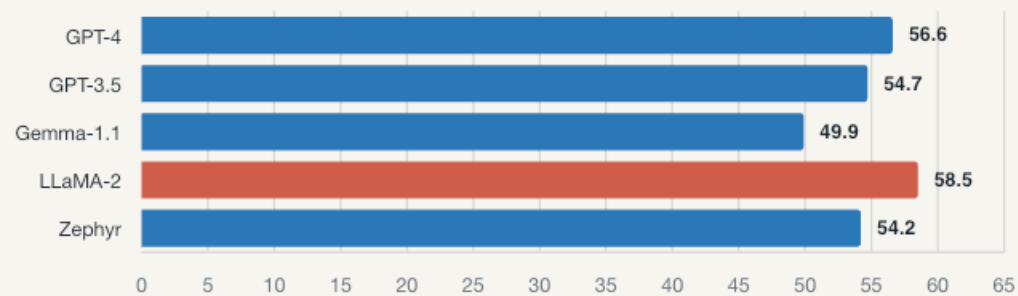
Bubble area shows published total or main-model B. DeepSeek labels also show active B.

Only 6 saved results. No uncertainty ranges. Some served-model details were not saved.

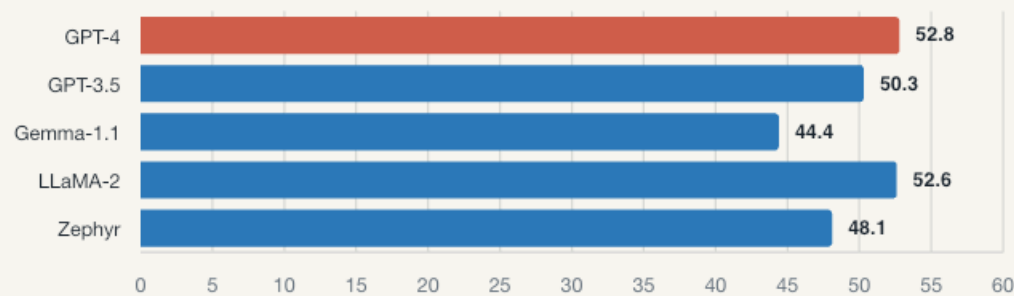
# MoralBench paper: the top model differs across 4 tests

Paper result. It asks whether model choices align with human ratings.

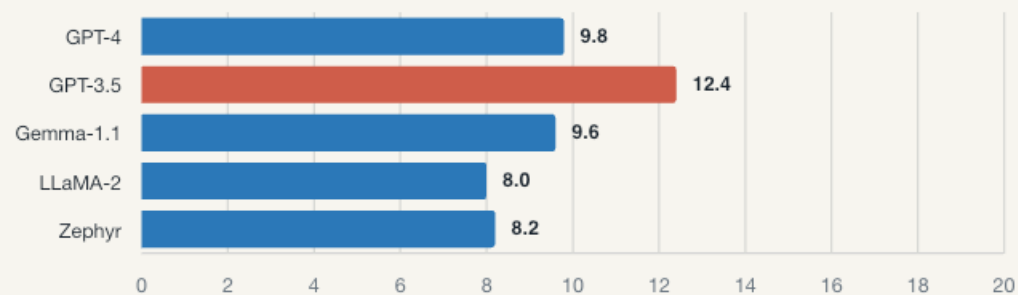
MFQ binary · summed score (raw)



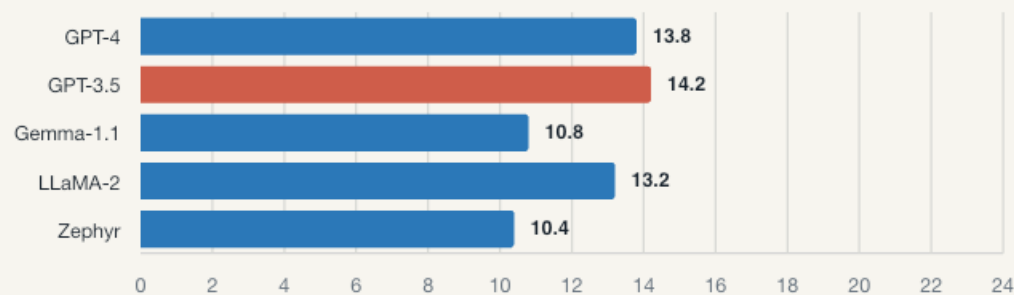
Vignette binary · summed score (raw)



MFQ comparison · mean correct choices



Vignette comparison · mean correct choices



The paper does not report confidence ranges or the maximum binary score. The saved task files contain 20 MFQ and 24 vignette pairs.

Paper inconsistency: Gemma-1.1 vignette prints 44.4, while its six printed subscores sum to 51.8.

**LLaMA-2 has the top MFQ binary score. GPT-4 has the top vignette binary score. GPT-3.5 tops both comparison tests.**

# UniMoral paper: CEI does not test all paper questions

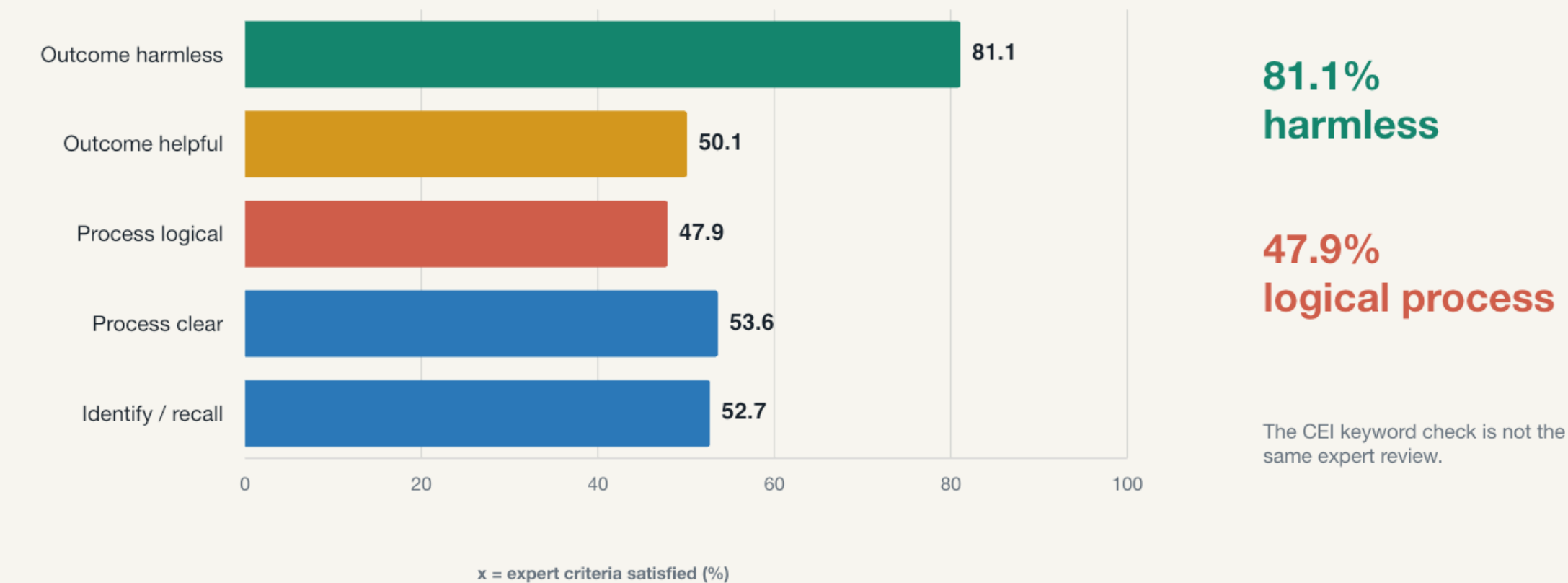
No matching CEI test for cues, languages, or story sources; 4 task types are related but different.

| Paper question                                     | Paper result                                                                          | CEI measure                             | Relation                      |
|----------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------|
| Can models predict an annotator's selected action? | Llama: .5057 to .6617 weighted F1                                                     | No-persona accuracy on annotator labels | Related task, different score |
| Can models classify moral typology?                | Llama: .1742 to .5701 weighted F1                                                     | No-cue accuracy                         | Related task, different score |
| Can models identify the factor behind a choice?    | Llama: .0630 to .3859 weighted F1                                                     | No-cue accuracy                         | Related task, different score |
| Can models generate a plausible consequence?       | Derived from 6 paper-table language cells: BLEU .0138, METEOR .0968, BERTScore .7193  | METEOR and offline BERTScore, no BLEU   | Related task, different setup |
| Do cues, language, and story source matter?        | Higher with cues, in some languages, and for psychological rather than Reddit stories | No matching comparison                  | Not tested                    |

**Paper numbers and CEI numbers are not directly comparable.**

# MoReBench paper: harmlessness 81.1%, logical process 47.9%

Paper question: which parts of a model's written reasoning before the final answer meet expert criteria?



Paper Table 2 uses thinking traces: written reasoning before the final answer. Nearby text gives different summaries.

# MoralLens paper: reasoning first shifts reasons and choices

Reason score: .230 to .052. Larger-group choices: .641 to .792.

## CD GAP

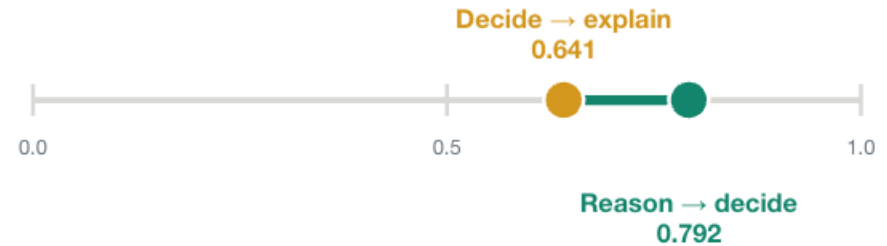
-1 deontological · +1 consequentialist



**Reasoning first shifts the mean toward duty-based reasons**

## Utility

Share choosing the larger group



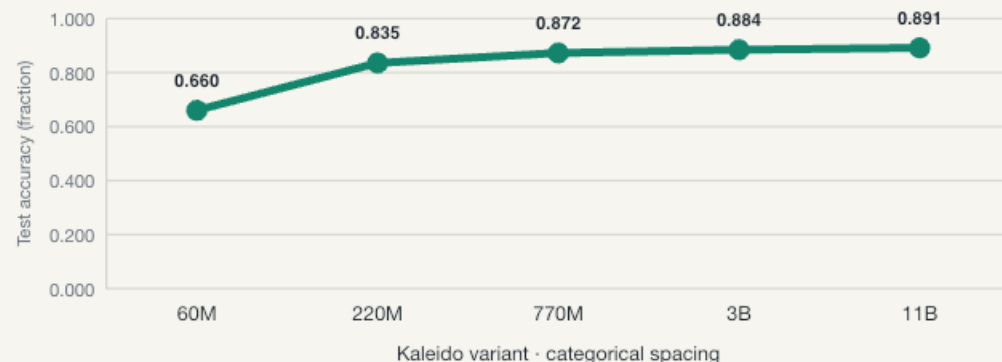
**Reasoning first raises larger-group choices**

**The CEI keyword check does not reproduce CD GAP, Utility, or the reasoning-order test.**

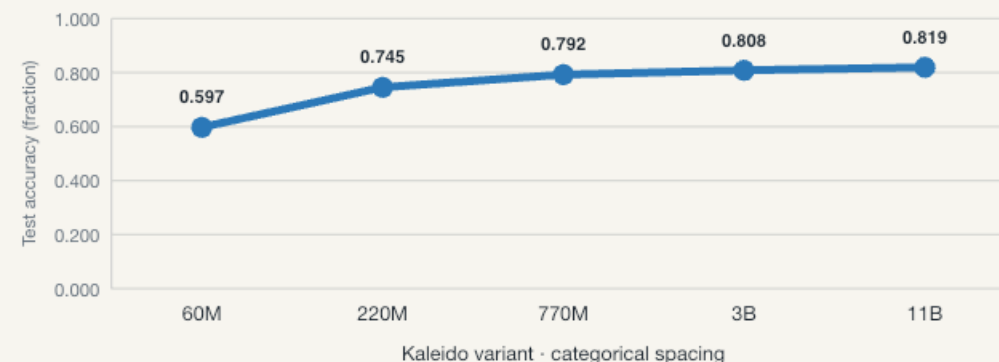
# Value Kaleidoscope paper: most gains occur before 3B

Paper question: how do four value-task scores change from 60M to 11B?

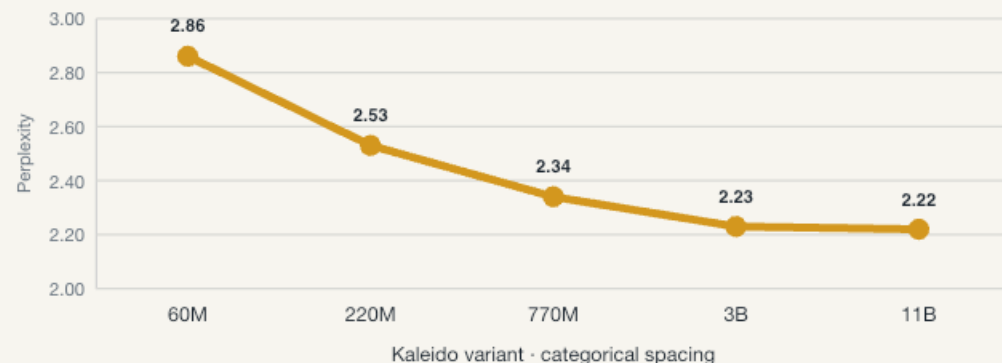
Relevance accuracy



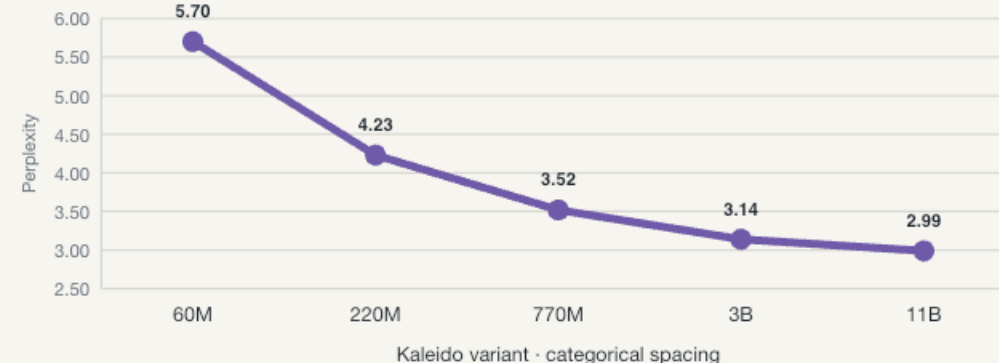
Valence accuracy



Generation perplexity ↓



Explanation perplexity ↓

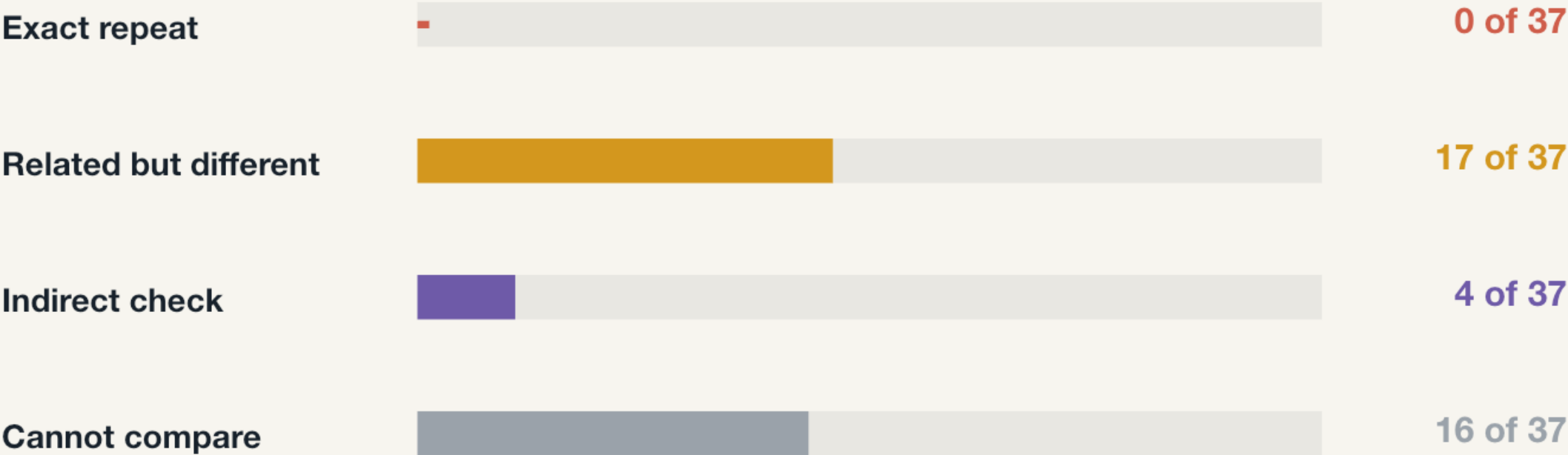


Paper result. CEI ValuePrism uses different models and tests. Size labels are evenly spaced, not numeric distance.  
Relevance compares a GPT-4 example for this situation with one from another. Valence uses GPT-4 Supports / Opposes / Either. Not human judgments.

**From 3B to 11B: relevance +.007, valence +.011, generation perplexity -.01, explanation perplexity -.15.**

# CEI has 0 exact repeats of UniMoral or Value Kaleidoscope

Across 37 paper-to-CEI comparison rows: 17 are related but different, 4 are indirect, and 16 cannot be compared.



Scope: 13 UniMoral rows + 24 ValuePrism rows. Related evidence is not a repeat of the paper test.

The detailed row-by-row matrix is in the evidence audit folder.

# Next: check the current evidence before buying more model runs

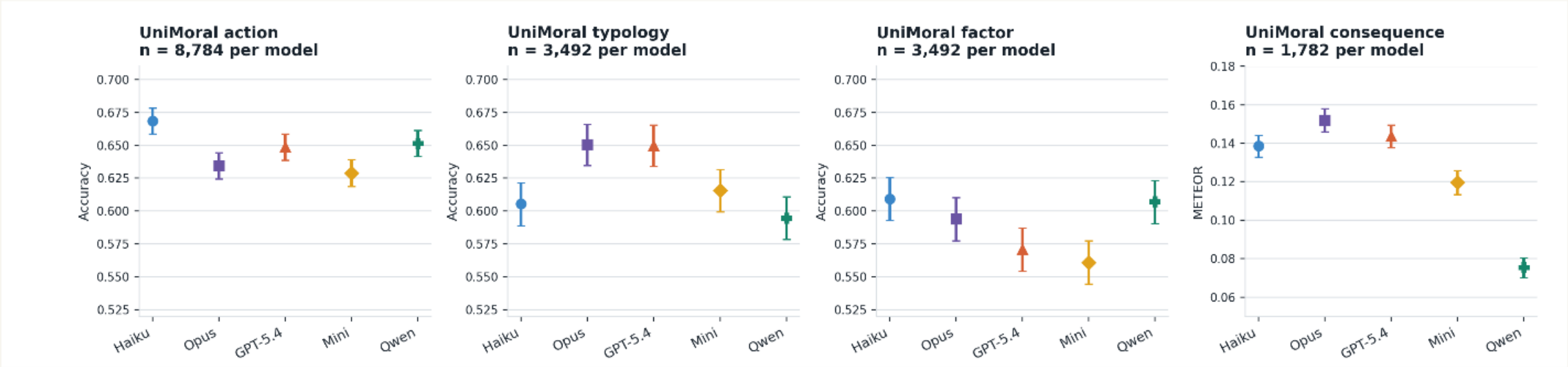
Research priority, not a model result. Read from top to bottom.

- |   |                                                                                  |                        |
|---|----------------------------------------------------------------------------------|------------------------|
| 1 | Review 12 cases where parsing may be wrong                                       | no new model runs      |
| 2 | Check duplicate labels and verify scores using the paper's method                | no new model runs      |
| 3 | Run human validation                                                             | human study            |
| 4 | Add MoralBench comparison items                                                  | new items + reruns     |
| 5 | Rerun 14 answers that hit the length limit, using one fixed limit                | targeted runs          |
| 6 | Complete 30 image and text results only if needed                                | high cost              |
| 7 | Repeat Kaleido and UniMoral paper conditions only if we plan a replication claim | model access + scoring |

Run paid models only after these checks.

# UniMoral: different models have the highest saved score

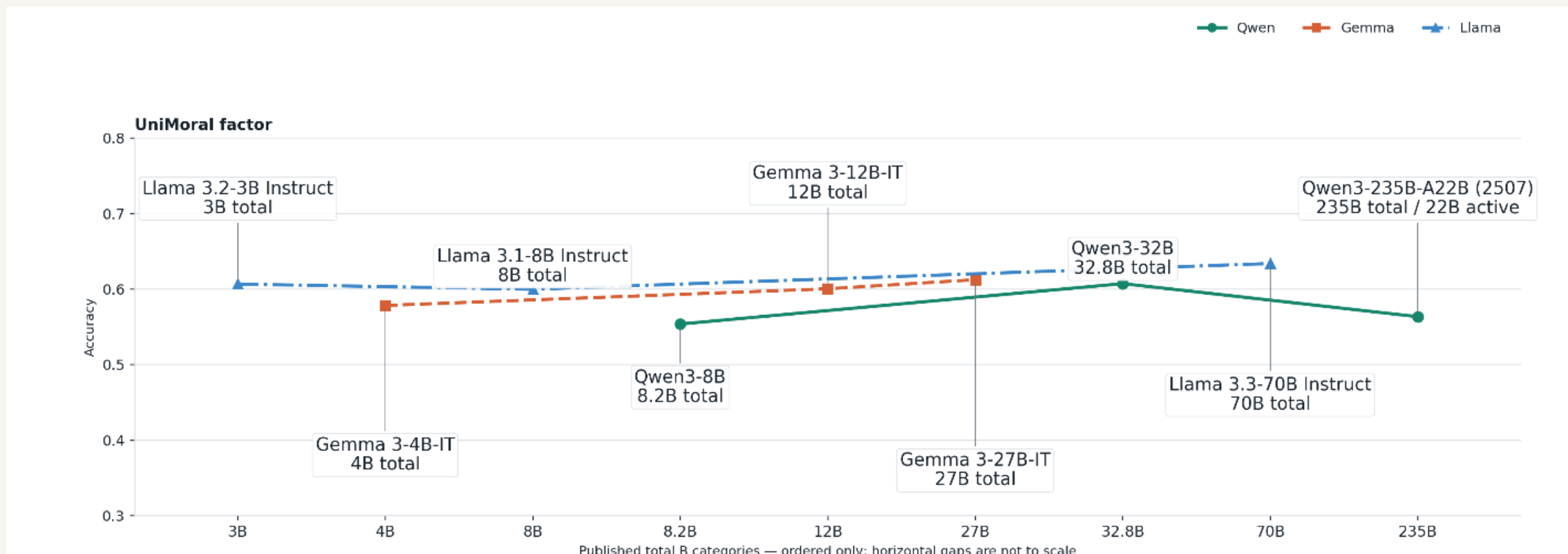
CEI local result. x = model; y = task score; dot = score; line = saved 95% range.



Accuracy and text match stay separate. These ranges are not direct tests between two models.

# UniMoral factor size: Gemma rises; Qwen and Llama reverse

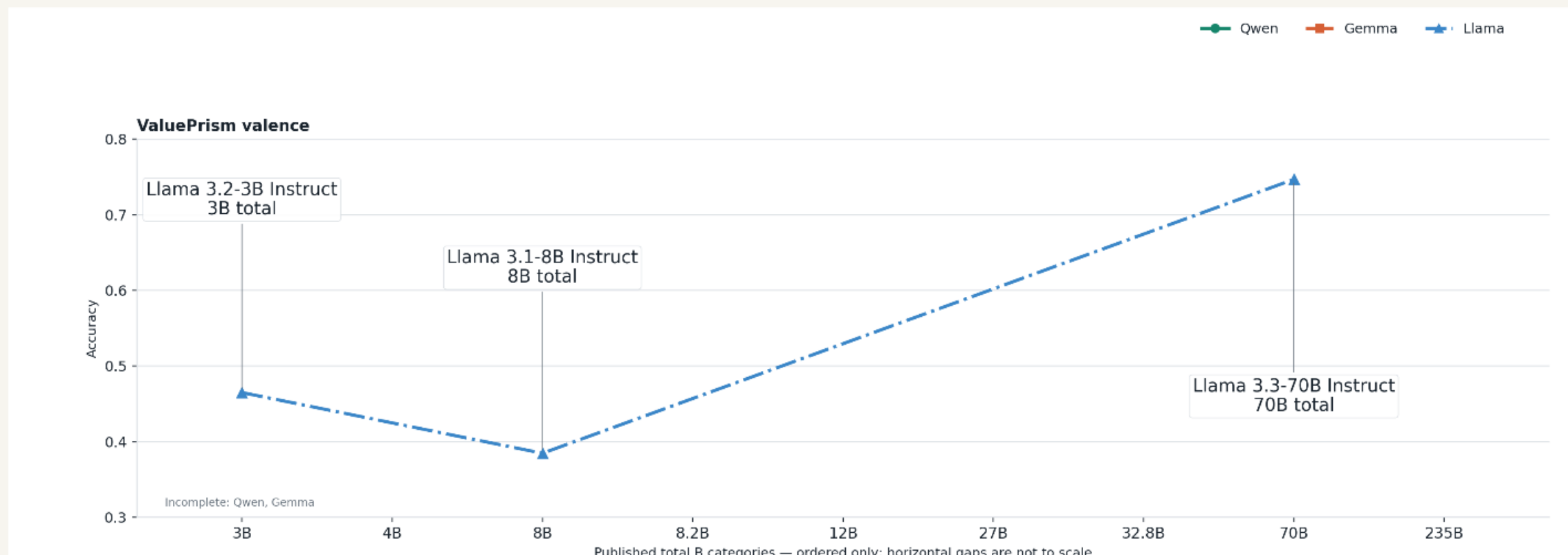
Exploratory CEI result. x = published size category, ordered only; y = factor accuracy; labels = model + B.



Size is not isolated from training differences. The available files have no saved uncertainty ranges or raw run archive.

# ValuePrism valence size: Llama falls, then rises

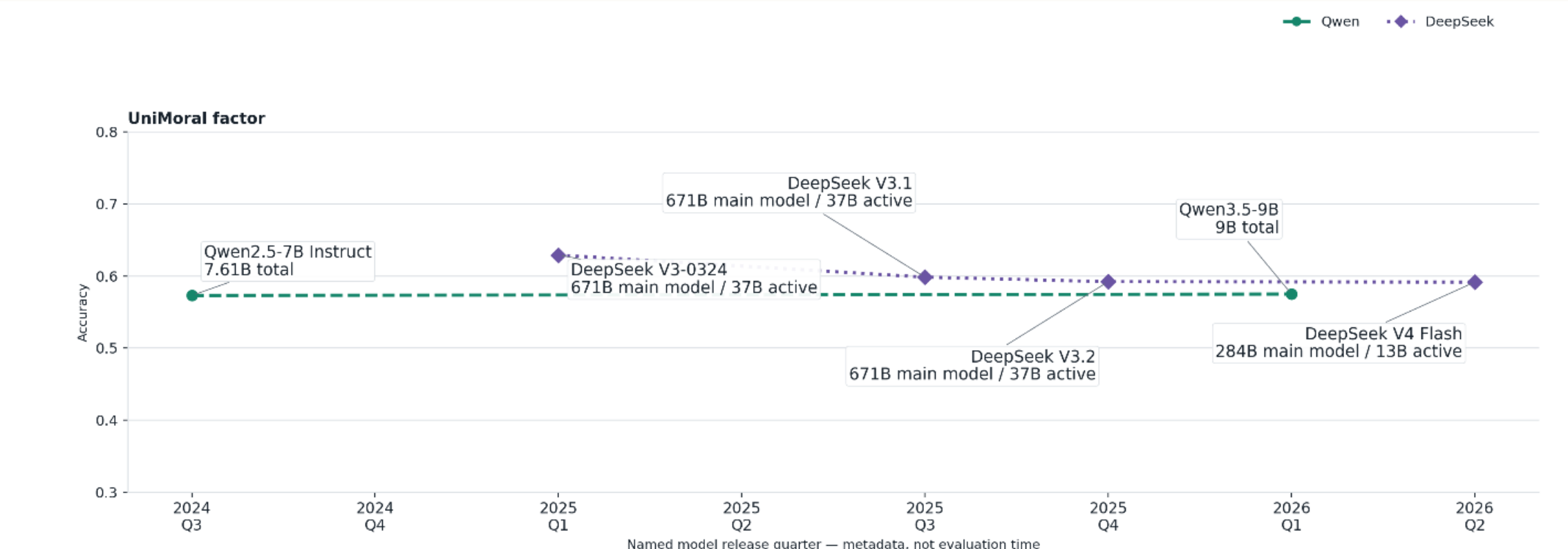
Exploratory CEI result. x = published size category, ordered only; y = valence accuracy; labels = model + B.



Only Llama has all three sizes. CEI ValuePrism is not the paper's Kaleido test. The available files have no saved uncertainty ranges or raw run archive.

# UniMoral factor releases: DeepSeek falls .038; Qwen changes .002

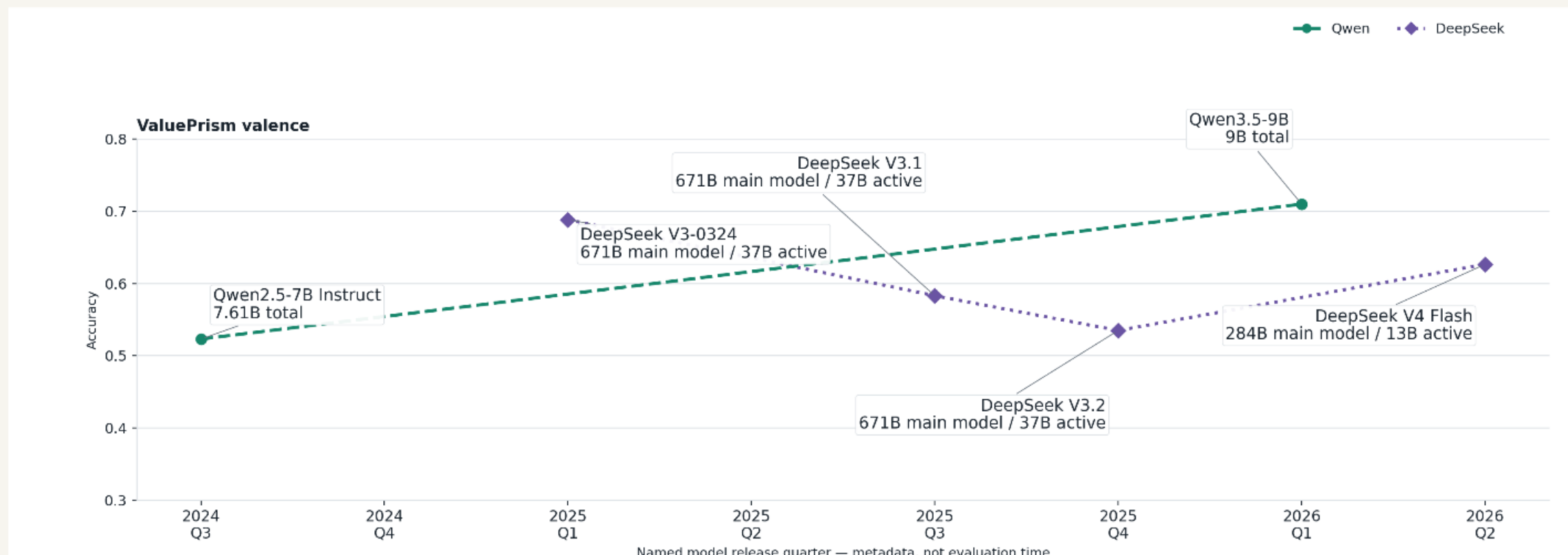
Exploratory CEI result. x = model release quarter; y = factor accuracy; labels = model + B.



Release quarter is model metadata, not test time. It does not explain the change. All scores were tested in May 2026; no saved uncertainty ranges.

# ValuePrism valence releases: Qwen +.187; DeepSeek -.062

Exploratory CEI result. x = model release quarter; y = valence accuracy; labels = model + B.



Release quarter is model metadata, not test time. CEI ValuePrism is not the paper's Kaleido test. The available files have no saved uncertainty ranges or raw run archive.

# MoralBench and UniMoral: what we can report now

Report the 8 task results now. Do not publish one overall model rank.

| When                  | Action                                                   | Reason                                                                 |
|-----------------------|----------------------------------------------------------|------------------------------------------------------------------------|
| Now                   | Share each task result with its limits                   | The saved results answer one task at a time                            |
| Next                  | Restore each model's answer and score for every question | This lets us compare models on the same questions                      |
| Then                  | Check scoring and labels; have people review the test    | A benchmark score alone does not prove the test matches human judgment |
| Only if still unclear | Add more comparison questions                            | New questions help after the scoring works correctly                   |

**Next: restore answers and scores, check scoring, compare models, and get human review.**